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Stat 330
DATA ANALYSIS: Irish Electricity

1 Summary

On January 1, 2010, there was a major change in the tariff structure on electricity consumption in Ireland. We thus seek to determine whether or not this change in electricity price will affect the patterns of power usage across the country. In essence, we seek to find if there is a difference in energy consumption patterns before versus after the policy change, and if, during the 2.5 months after the change, the pattern of energy consumption return to the pattern before the policy change. We looked at electricity consumption (in kilowatt-hour (kWh)) averaged over 151 randomly sampled households in Ireland for the time between November 15, 2009, to March 15, 2010. We have manipulated the data so that for each day in the dataset, we have the following:

Table 1:

Variable Name	Description
MinTemp	Low Temperature in ◦C
Day	Day of the Week
near_holiday	If the given day is a holiday* or the three days before/after
month	Month (November-March)
tariff	If the given day is before/after Jan 1
Energy	Energy Consumption in kWh

*for our purposes we use major holidays which include Christmas, New Year's Day, and St. Patrick's Day

Using our final model, we were able to find that the tariff very significantly affected energy usage in Ireland. We also found that minimum temperature was a significant factor in energy consumption before, but not after the tariff. In essence, relationship between temperature and power usage was effected by the tariff.

2 Data & Exploratory Analysis

Preliminarily, we found that the initial effect of the tariff was very stark. Even at the most extreme, we never saw post-tariff energy consumption reach even the minimum amount of consumption that was had pre-tariff. It appears possible that minimum temperature may have continued to affect energy usage post-tariff as it did in November and December, but the effect is far less prevalent.

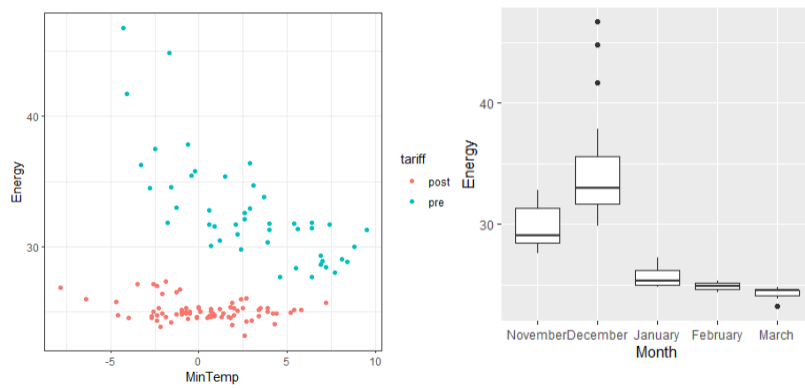

Figure 1: Energy consumption plotted against minimum temperature and stratified by pre- and post-tariff months (left) and energy consumption by pre- and post- tariff months (right).

From Figure 1, we see a much larger spread of values pre-tariff than post-tariff, thus we wanted to look at more variables than just temperature to see what other energy usage patterns might have changed. From Figure 2, we determined that there may have been a trend by weekday, but after looking at preliminary models, it was shown that it did not add predictive power and no significant trend existed.

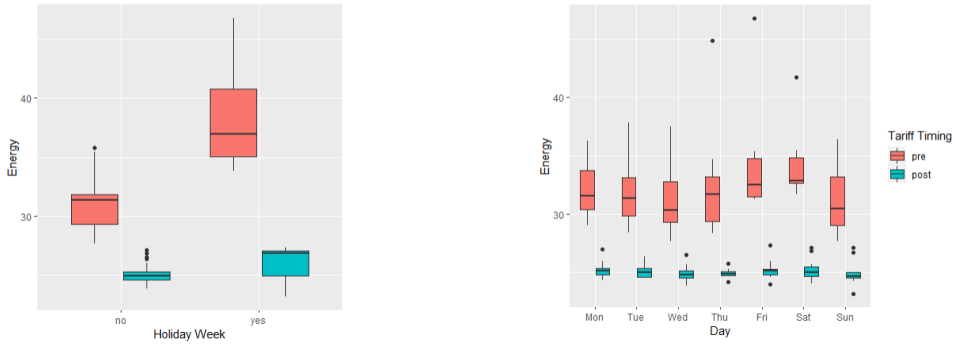

Figure 2: Energy consumption across categorical variables and stratified by pre- and post-tariff: holiday proximity (left), day of the week (right).

We decided to test linearity, normality, and homoscedasticity to determine what kind of models would be appropriate to use. We also checked for outliers. Using a Cook's Distance analysis, we found that Christmas, Christmas Eve, and Boxing Day were all potential outliers. To attempt to combat this, we decided to make a the near_holiday variable as explained in Table 1. In preliminary models near_holiday seemed to be a good variable to look at, but when running the same model with the three days removed, it ceased to help in prediction, so we opted to keep the Dec. 24, 25, 26 points out of our data set.

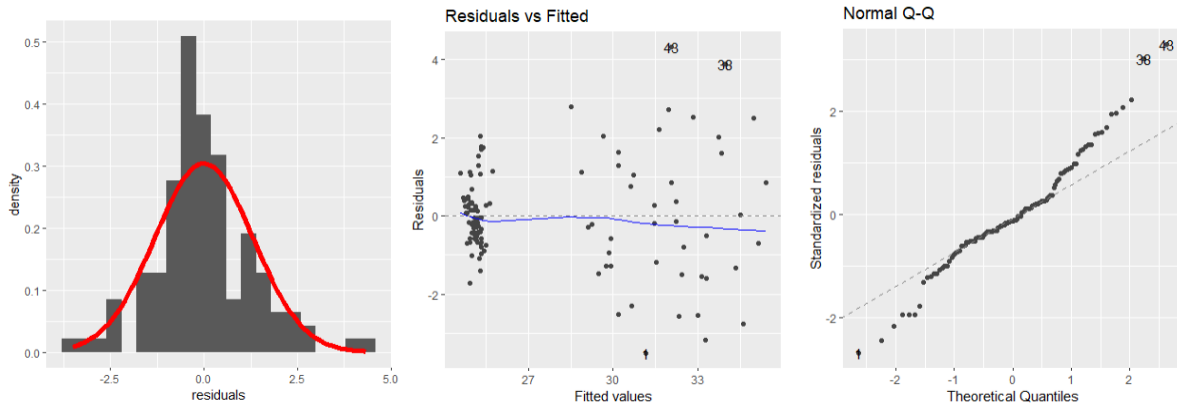

Figure 3: Two of the four plots used for assessing linearity, normality of residuals and homoscedasticity assumptions: a histogram of residuals for normality (left), a residuals vs. fitted values plot for linearity and homoscedasticity (center), and a Q-Q plot for for normality (right).

Our residual vs. fitted plot shows homoscedasticity within the two tariff groups and, when removing outliers, also between groups. From that plot we may also assume linearity. Our Q-Q plot and histogram together demonstrate relative normality, but it maintaining the normality assumption may lead to problems in the model.

3 Methods and Models

We considered a total of 13 models: polynomial models from degree one to three with and without interaction allowed, spline models from degree one to three with and without interaction allowed, and a standard linear model with an interaction between tariff and MinTemp. Using the AIC and BIC values for each of the 13 models, we were able to determine that the best model was the below linear model with interaction.

$$y_i = \beta_0 + I(\text{tariff}_i = \text{pre})*\beta_1 + \text{MinTemp}_i*\beta_2 + (\text{MinTemp}_i \times I(\text{tariff}_i = \text{pre}))*\beta_3$$

Although, the polynomial model with interaction has the same AIC and BIC as our chosen model, the models did have different coefficients, so we opted for the simpler model for the sake of higher interpretability.

4 Results and Discussion

As mentioned previously our model was chosen for it's greater level of interpretability compared to all other models we tested as well as the fact that it yielded the most optimal AIC/BIC values. Table 2 shows the model summary.

Table 2:

```
lm(formula = Energy ~ MinTemp + tariff + tariff * MinTemp, data = elec)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.5032 -0.6737 -0.1614  0.4839  4.3286 

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)    
(Intercept)        25.12794    0.15539 161.714  < 2e-16 ***
MinTemp            -0.07578    0.05443  -1.392    0.167    
tariffpre           8.51001    0.31846  26.723  < 2e-16 ***
MinTemp:tariffpre  -0.46479    0.07969  -5.832  5.2e-08 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.329 on 114 degrees of freedom
Multiple R-squared:  0.875,	Adjusted R-squared:  0.8717 
F-statistic: 266.1 on 3 and 114 DF,  p-value: < 2.2e-16
```

We see from the summary that temperature alone is not a significant predictor for energy consumption, but the interaction between temperature and tariff is. This gives an indication that pre-tariff, energy consumption was tied to temperature, but this cannot be said to be the case post-tariff based on our model. Figure 4 also shows a highly negative correlation between temperature and energy usage pre-tariff, and a far less negative correlation post-tariff.

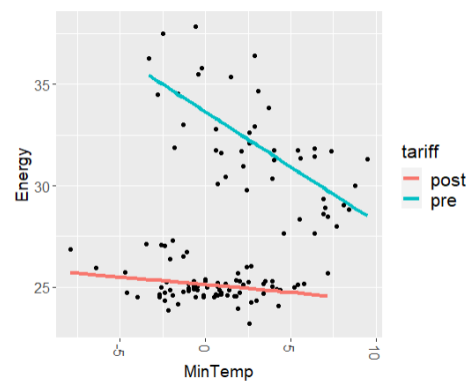

Figure 4: Visual plot of the chosen model against our cleaned data

As discussed before, holiday times may be a factor worth exploring. However, with only two full holiday weeks in this data set it is hard to make any conclusions especially since the two holidays are exactly one week apart and the week of New Year’s being split between the tariff groups. It would be very useful to have data for the remainder of the 2009-10 winter, to get the full data of St. Patrick’s Day as well as the data for November and December of 2010.

5 Conclusion

In this analysis, we have shown that after the imposition of the tariff, causing a rise in energy prices, energy consumption decreased noticeably and significantly. Not only this, but, based on our model, after the tariff we did not see temperature correlate with energy consumption in the same way. We could not conclusive say anything about holiday weeks and their relationship with energy consumption because of insufficient data, but because of high usage around Christmas, we may hypothesize that with further data and remodeling we could see great significance, even post-tariff.